

```
In [2]: import pandas as pd
```

```
In [3]: dataset=pd.read_csv("Salary_dataset.csv")
```

```
In [4]: dataset
```

Out[4]:

	Unnamed: 0	YearsExperience	Salary
0	0	1.2	39344.0
1	1	1.4	46206.0
2	2	1.6	37732.0
3	3	2.1	43526.0
4	4	2.3	39892.0
5	5	3.0	56643.0
6	6	3.1	60151.0
7	7	3.3	54446.0
8	8	3.3	64446.0
9	9	3.8	57190.0
10	10	4.0	63219.0
11	11	4.1	55795.0
12	12	4.1	56958.0
13	13	4.2	57082.0
14	14	4.6	61112.0
15	15	5.0	67939.0
16	16	5.2	66030.0
17	17	5.4	83089.0
18	18	6.0	81364.0
19	19	6.1	93941.0
20	20	6.9	91739.0
21	21	7.2	98274.0
22	22	8.0	101303.0
23	23	8.3	113813.0
24	24	8.8	109432.0
25	25	9.1	105583.0
26	26	9.6	116970.0
27	27	9.7	112636.0
28	28	10.4	122392.0
29	29	10.6	121873.0

In [5]: dataset = dataset.drop(columns=['Unnamed: 0'])

In [6]: dataset

Out[6]:

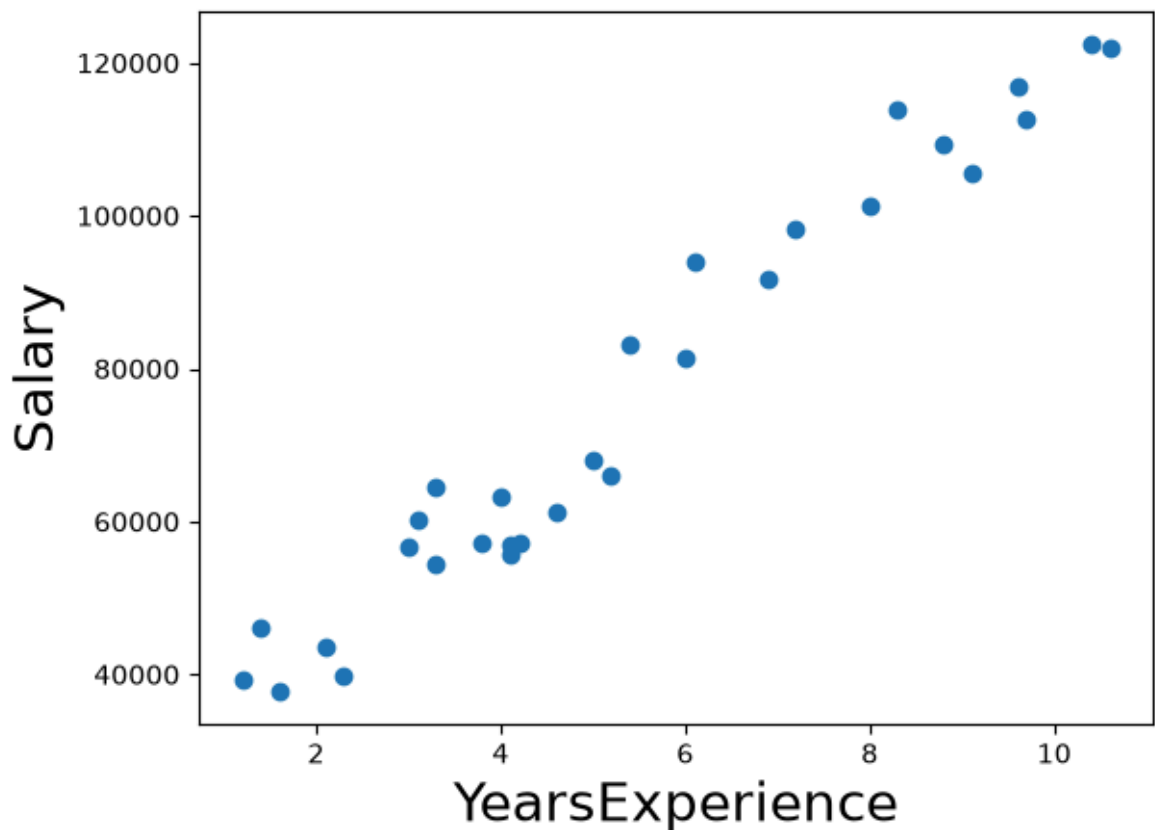
	YearsExperience	Salary
0	1.2	39344.0
1	1.4	46206.0
2	1.6	37732.0
3	2.1	43526.0
4	2.3	39892.0
5	3.0	56643.0
6	3.1	60151.0
7	3.3	54446.0
8	3.3	64446.0
9	3.8	57190.0
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20	6.9	91739.0
21	7.2	98274.0
22	8.0	101303.0
23	8.3	113813.0
24	8.8	109432.0
25	9.1	105583.0
26	9.6	116970.0
27	9.7	112636.0
28	10.4	122392.0
29	10.6	121873.0

```
In [7]: dataset.isnull().sum()
```

```
Out[7]: YearsExperience    0  
Salary                  0  
dtype: int64
```

```
In [8]: x=dataset[["YearsExperience"]]  
y=dataset[["Salary"]]
```

```
In [9]: import matplotlib.pyplot as plt  
plt.scatter(x,y)  
plt.xlabel('YearsExperience', fontsize=20)  
plt.ylabel('Salary', fontsize=20)  
plt.show()
```



```
In [10]: from sklearn.model_selection import train_test_split  
x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.2,random_s
```

```
In [11]: from sklearn.linear_model import LinearRegression  
regressor=LinearRegression()  
regressor.fit(x_train,y_train)
```

```
Out[11]: ▾ LinearRegression ⓘ ?
```

▸ Parameters

▸ Fitted attributes

```
In [12]: weight = regressor.coef_  
print("Weight of the model={}".format(weight))
```

```
bais = regressor.intercept_  
print("Intercept of the model={}".format(bais))
```

Weight of the model=[[9504.98248109]]
Intercept of the model=[24425.80702317]

```
In [13]: y_pred=regressor.predict(x_test)  
y_pred
```

```
Out[13]: array([[ 71950.71942864],  
                [ 55792.24921078],  
                [ 53891.25271456],  
                [115673.63884167],  
                [ 44386.27023347],  
                [ 63396.23519566]])
```

```
In [14]: x_test
```

```
Out[14]:      YearsExperience
```

15	5.0
8	3.3
6	3.1
26	9.6
3	2.1
11	4.1

```
In [15]: y_test
```

```
Out[15]:      Salary
```

15	67939.0
8	64446.0
6	60151.0
26	116970.0
3	43526.0
11	55795.0

```
In [16]: y_pred
```

```
Out[16]: array([[ 71950.71942864],  
                [ 55792.24921078],  
                [ 53891.25271456],  
                [115673.63884167],  
                [ 44386.27023347],  
                [ 63396.23519566]])
```

```
In [17]: import pandas as pd  
results = pd.DataFrame({  
    'YearsExperience': x_test.iloc[:, 0].values,
```

```

    'Actual_Salary': y_test.iloc[:, 0].values,
    'Predicted_Salary': y_pred.flatten()
})
print(results)

```

	YearsExperience	Actual_Salary	Predicted_Salary
0	5.0	67939.0	71950.719429
1	3.3	64446.0	55792.249211
2	3.1	60151.0	53891.252715
3	9.6	116970.0	115673.638842
4	2.1	43526.0	44386.270233
5	4.1	55795.0	63396.235196

```

In [18]: from sklearn.metrics import r2_score
ab=r2_score(y_test,y_pred)*100
ab

```

Out[18]: 94.08827284776221

```

In [19]: import pickle
filename = 'finalized_model.sav'
pickle.dump(regressor, open(filename, 'wb'))

```

```

In [20]: loaded_model = pickle.load(open('finalized_model.sav', 'rb'))
result = loaded_model.predict([[15]])
print(result)

```

[[167000.54423958]]

C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(

```

In [21]: prediction_input = int(input("Enter the Prediction input value:"))
if prediction_input > 60:
    print("Person doesn't exist anymore")
else:
    Future_Prediction = regressor.predict([[prediction_input]])
    print("Future_Prediction={}".format(Future_Prediction))

```

Future_Prediction=[[547199.84348331]]

C:\Users\goushik lap\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(

In []:

In []:

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